

**PRODUCTION OF BIODIESEL FROM *THEVETIA PERUVIANA* SEED
OIL USING NaOH AS CATALST**

BY

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CERTIFICATION

This is to certify that I, **OLANIPEKUN DEBORAH ADETUTU** with the **Matriculation Number CHM/2021/1054** of the Department of Chemistry, Faculty of Physical Sciences, Federal University Oye-Ekiti, Ekiti State have successfully written and submitted my project report.

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DEDICATION

I dedicate this project result to my late father, Dr. J.A Olanipekun who has supported me by giving me the best education journey from elementary to university.

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First and Foremost, All Glory and Adoration be to God for granting me the strength, wisdom, and opportunity to successfully complete this project write-up.

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ABSTRACT

Biodiesel is one of the many alternative fuels that can replicate diesel oil's combustion characteristics and be readily modified for use in current engine technologies. One viable alternative to conventional diesel fuel that has positive effects on the environment is biodiesel fuel made from vegetable oils (both edible and non-edible) or animal fats. The production and characterization of biodiesel from *Thevetia peruviana* seed oil using sodium hydroxide (NaOH) and potassium hydroxide (KOH) catalysts were investigated. In order to evaluate *Thevetia peruviana* seed oil's viability as a non-edible feedstock for biodiesel production, its physicochemical characteristics were assessed. 1g of NaOH and 0.5g of KOH were used as catalysts in the base-catalyzed transesterification of 15.82g of methanol to create biodiesel. Fourier Transform Infrared (FTIR) spectroscopy was used to analyze the conversion of the triglycerides in the seed oil to fatty acid methyl esters (FAME) and identify the functional groups present in the resulting biodiesel samples. The FTIR spectra of the biodiesel made with both catalysts and the raw *Thevetia peruviana* oil were compared. The synthesis of biodiesel was confirmed by the biodiesel samples' distinctive absorption bands linked to ester functional groups and aliphatic C–H stretching. The KOH-catalyzed biodiesel, on the other hand, displayed more pronounced spectrum evidence of transesterification, such as the absence of the raw oil's 1103 cm^{-1} peak and the presence of a distinctive ester carbonyl (C=O) absorption band. By contrast, the biodiesel catalyzed by NaOH did not exhibit the same level of spectrum change. These results imply that, in the conditions examined, KOH exhibited superior catalytic performance compared to NaOH. Overall, the findings show that catalyst selection has a major impact on the transesterification process' efficiency and highlight the potential of *Thevetia peruviana* seed oil as a renewable, non-edible feedstock for biodiesel synthesis.

CHAPTER ONE

INTRODUCTION

1.0 BACKGROUND TO THE STUDY

The global energy sector predominantly relies on natural gas, coal, and petroleum resources to fulfill its energy requirements (Meher *et al.*, 2006). Nigeria, which exports more than 70% of its crude oil production, has historically been an energy-deficient nation. For the nation to continue growing, petroleum products must be imported. Nigeria's industrial economy depends heavily on diesel fuel. Heavy vehicles, municipal transport buses, electric generators, farm equipment, and other devices use this fuel (Anjana, 2000). However, diesel engines can release a variety of pollutants into the environment that can harm the ecosystem and put human health at risk (Antolin *et al.*, 2002). Therefore, it is imperative that the globe broaden its focus to include new energy sources.

Biofuel, or fuel derived from renewable resources, is a relatively new alternative that is currently experiencing a global surge. According to Nwafor and Nwafor (2000) and Ezeanyananso *et al.* (2010), biofuels are a good fit for decentralized development, meaning they may be used to meet the needs for social and economic advancement, particularly in rural communities where fossil fuels may be costly or difficult to get.

Biodiesel is one of the many alternative fuels that can replicate diesel oil's combustion characteristics and be readily modified for use in current engine technologies. One viable alternative to conventional diesel fuel that has positive effects on the environment is biodiesel fuel made from vegetable oils (both edible and non-edible) or animal fats. Biodiesel is an alcoholic ester of various fatty acids, also known as FAME (Fatty Acid Methyl Esters). The major components of plant oils and animal fats are triacylglycerol (TAGs); the esters of fatty

acids and glycerol. Because biodiesel is a domestic, renewable resource with an eco-friendly emission profile, easily accessible, and biodegradable, it is advised to be used in place of petroleum diesel (Zhang *et al.*, 2003).

Due to the diesel crisis brought on by the Organization of Petroleum Exporting Countries' (OPEC) drop in petroleum production and the ensuing price increase, study and use of biodiesel fuel as an alternative began in the 1980s. The biodiesel made from locally accessible resources has a lot of potential for use in Nigeria in the future since it will help achieve much-needed energy security and is ecologically harmless, which will help it comply with higher emission standards (Ezeanyananso, 2010). The primary feature of biodiesel is that its viscosity, cetane number, and energy content are comparable to those of regular diesel.

Additionally, it can be used in any ratio with diesel generated from petroleum. Consequently, one of the most widely used biofuels worldwide is biodiesel. Free fatty acids (FFA) are transesterified or esterified with short chain alcohol (methyl or ethyl alcohol) to produce biodiesel, a renewable fuel that is less polluting than diesel and biodegradable.

1.1 PROBLEM STATEMENT

Since edible oil competes with humans for food sources and there is a significant gap between supply and demand, using edible oil to make biodiesel is not practical. In order to produce biodiesel, alternative non-edible food sources must be investigated.

Non-edible vegetable oils, which are mostly produced by trees and bushes that grow seeds, can offer a substitute that has no conflicting food applications. However, the high free fatty acid (FFA) content of crude non-edible oils has an impact on the production and capital cost of biodiesel. By employing esterification and saponification techniques to lower the free fatty acid content (FFA), these oils can be pre-refined.

One of the main obstacles to the commercialization of biodiesel is its high cost; depending on the feedstock used, it can be 150% more expensive than petroleum-based diesel. The cost of the feedstock, solvent, and crude oil refining process accounts for between 70 and 95 percent of the entire cost of producing biodiesel (Hass *et al.*, 2006; Umer and Farooq, 2008).

1.2 AIM AND OBJECTIVES OF THE STUDY

1.2.1 Aim of the study

The aim of this project was to study the production of biodiesel from the transesterification of *Thevetia peruviana* Seed Oil using NaOH as the catalyst.

1.2.2 Objectives of the study

The objectives of this study were to:

- i. produce biodiesel from *Thevetia peruviana* seed oil by transesterification;
- ii. to characterize the biodiesel using FTIR;
- iii. to compare the result with product produced from using KOH as catalyst and TVPO oil.

1.3 JUSTIFICATION OF THE STUDY

Diesel fuels made from vegetable oils virtually contain no Sulphur, emit no greenhouse gases, particularly CO₂, are easy to store, and have superior lubricating qualities. Additionally, burning vegetable oils causes trees to absorb more carbon dioxide from the atmosphere (Erhan and Sharma, 2006). Therefore, diesel fuels essentially contribute to reducing the atmospheric concentration of carbon dioxide. This breakthrough has led to a resurgence of interest in using vegetable oils to produce biodiesel fuels (Kim *et al.*, 2004).

Even for major oil exporting nations like Nigeria, replacing diesel oil with domestically produced renewable fuels will result in greater foreign exchange savings. Thus, this type of project can be used by underdeveloped nations to address their economic issues as well as their environmental issues (Ezeanyananso *et al.*, 2010). It is necessary to produce biodiesel with a less expensive reagent, which lowers capital and manufacturing costs. Additionally, before employing biodiesels, diesel fuels, and their blends in a diesel engine, more research is required about their fuel characteristics (Hass *et al.*, 2006).

CHAPTER TWO

LITERATURE REVIEW

2.1 OVERVIEW OF BIODIESEL

The fact that biodiesel emits fewer dangerous gases, such as carbon monoxide, Sulphur dioxide, and aromatic hydrocarbons, is a significant advantage over petroleum fossil. According to Balat (2007) and ASTM (2015), biodiesel requires no modification to be used in conventional diesel engines and can be blended with fossil fuel at any desired ratio. Transesterification, which uses lipid and low carbon alcohol as raw materials in the presence of a catalyst, is the most popular process for producing biodiesel (Bateni *et al.*, 2017; Buchori *et al.*, 2016; Orege *et al.*, 2022). Oil and alcohol undergo a transesterification reaction to produce biodiesel, an alternative fuel.

FAME (Fatty Acid Methyl Esters), another name for biodiesel, is an alcoholic ester of several fatty acids. Triacylglycerol (TAGs), fatty acid esters, and glycerol are the main constituents of plant and animal fats (Keneni & Marchetti, 2017; Bateni *et al.*, 2017). Environmental benefits of biodiesel include low emissions of carbon dioxide and monoxide, low toxicity, and biodegradability. The primary feature of biodiesel is that its viscosity, cetane number, and energy content are comparable to those of regular diesel (Mahajan, 2011; Balat, 2007). Additionally, diesel derived from petroleum can be blended in any ratio (Balat, 2007; Mofijur *et al.*, 2015).

Consequently, one of the most widely used biofuels worldwide is biodiesel (Orege *et al.*, 2022; Das, 2023). Triglycerides transesterification or esterification of free fatty acids (FFA) with short chain alcohol (methyl or ethyl alcohol) produces biodiesel, a renewable fuel that is less polluting than diesel and biodegradable (Bateni *et al.*, 2017; Malabadi *et al.*, 2023; Orege *et al.*, 2022).

2.1.1. History and Development of Biodiesel

When Dr. Rudolf Diesel (1858–1913) created the first diesel engine in 1895 with the goal of using vegetable oils, the idea of biodiesel as an engine fuel was born (Ma and Hanna, 1999). At the 1900 World Exhibition in Paris, he first demonstrated its concept using peanut vegetable oil. Diesel stated in 1912 that "the use of vegetable oils as engine fuel may seem insignificant today." However, these oils could eventually grow to be just as significant as current petroleum and coal tar products. Rudolf Diesel's prophecy has come to pass. Petroleum is understood to be a limited resource, and as its reserves diminish, its price tends to rise exponentially (Conceñ *et al.*, 2005).

Dr. Diesel's engine has been modified to run on the toxic petroleum fuel that we now refer to as "diesel" since his tragic death in 1913. However, his invention and agricultural theories lay the groundwork for a society powered by clean, renewable, locally produced fuel (Ma & Hanna, 1999; Schuchardt *et al.*, 1998). Methyl rapeseed oil was initially utilized for 1000 hours in engines by American scientists in 1983. A renewable fatty acid monoester is what is known as biodiesel.

The use of animal or plant fatty acid monoesters, such as fatty acid methyl ester, fatty acid ethyl ester, and fatty acid propyl ester, in place of diesel fuel combustion was investigated by researchers in the United States and Germany in 1984. Due to its renewable source and lower pollution, nations all over the world are now resuming the use of this fuel.

2.2.2 Sources of Biodiesel Feedstocks

Microalgae, animal fat, vegetable oil, algae, oils from fungi, terpenes, latex, edible waste oil, and excess cooking oil from residential and commercial operations are some of the sources of biodiesel (Balat, 2007; Orege *et al.*, 2022; Keneni & Marchetti, 2017; Malabadi *et al.*, 2023). Fish, pork lard, beef tallow, and poultry are examples of animal fats. Edible vegetable oils,

including those from sorghum, sunflower, barley, peanuts, oats, and coconuts, have been used for a long time. Biodiesel is made from about 17% of the world's total grain production. However, because to their conflict with food supplydemand, nonedible vegetable oils are preferable feedstock for biofuel synthesis (Balat, 2007; Keneni & Marchetti, 2017).

Low production costs and a big production scale are two requirements that biodiesel feedstock must meet. *Raphanus sativus*, *Raphanus raphanistrum*, *Prunus cerasoides*, and *Lepidium perfoliatum* are examples of non-edible oil-producing seed plants that could serve as inexpensive feedstock. These resources are perfect for addressing food insecurity. Indigenous non-edible feedstock has been found to be important in addressing the food supply debate and reducing air pollution. A significant industrial and nutritional crop, industrial hemp is a variation of the *Cannabis sativa* plant (*Cannabis sativa* Linn) (Hasnain *et al.*, 2023; Malabadi *et al.*, 2023).

Through the process of transesterification, hemp seed oil can be utilized to create biodiesel (Taufiq-Yap *et al.*, 2011). A feasible feedstock alternative for the manufacturing of biodiesel is oil derived from hemp seeds. The fact that biodiesel doesn't require any changes to the diesel engine is another significant benefit. Diesel fuel can be replaced with carbon-neutral hemp biodiesel. Photosynthesis reabsorbs the carbon dioxide emissions discharged into the atmosphere during the combustion of biodiesel (Parvez *et al.*, 2021; Malabadi *et al.*, 2023). Waste gathered on beaches can be addressed by using macro or microalgae as a raw material for biodiesel production. Crushing and solvent extraction are required steps in the synthesis of algal biodiesel. However, costly separation procedures may be required when using a solvent for extraction (Cancela *et al.*, 2012; Malabadi *et al.*, 2023).

Transesterification, pyrolysis, microemulsions, direct usage, and blending of raw oils are the four ways to produce biodiesel, a mixture of fatty acid methyl esters (FAME). The high viscosity and poor volatility of vegetable and animal fats make direct use unsuitable. The

cleavage of vegetable oils into smaller molecules utilizing thermal energy over catalysts is known as pyrolysis (Cancela *et al.*, 2012; Maceiras *et al.*, 2011). Nevertheless, it is a difficult procedure that yields unprofitable byproducts. Alcohol-based microemulsions have been developed to address the issue of vegetable oils' high viscosity. The transesterification process with a catalyst is the most often utilized technique. Transesterification is the process by which an alcohol and an oil react to produce glycerol, and catalysts might be present or absent during this reaction. According to Cancela *et al.* (2012), Balat (2007), and Malabadi *et al.* (2023), the reaction is reversible. In order to move the equilibrium to the product side, an excess of alcohol must be used. Three phases make up the reaction: (1) triglycerides are converted to diglycerides; (2) higher glycerides are converted to lower glycerides; and (3) glycerol is produced.

Waste oils, animal fats, and edible (first-generation) and non-edible vegetable oils (second-generation feedstocks) are possible feedstocks for the synthesis of biodiesel. First-generation biofuels, which are typically made from edible oils like soybeans, palm oil, sunflower, safflower, rapeseed, coconut, and peanut, are closely associated with biomass that is typically edible (Balat, 2007; Keneni & Marchetti, 2017; Malabadi *et al.*, 2023). From lignocellulosic materials to municipal solid wastes, a variety of feedstocks are used to manufacture second-generation biofuels. Algae, which are regarded as new non-edible oil sources because of their high oil content, quick biomass production, and potential for CO₂ usage, are related to third-generation biofuels.

In order to address the issues related to the ongoing use of edible oils for biodiesel production, non-edible biodiesel feedstocks have drawn a lot of interest.

2.2 THEVETIA PERUVIANA (YELLOW OLEANDER)

2.2.1 Botanical Description

Thevetia peruviana is a member of the Apocynaceae family of small ornamental shrubs. It is found throughout India and is also known as yellow oleander and pilli kaner due to its yellow blossoms (Pandya *et al.*, 2013). Kaner, digoxin, fortunate nut, milk bush, and yellow oleander are some other names for it.

This plant, which originated in Mexico, South America, and Central America, is now a tropical, evergreen decorative tree or bush that can grow up to 10 or 15 feet in height. The plant has diuretic, cardiotonic, and oedema-curing properties (Pandya *et al.*, 2013; Essiett and Udofa, 2014). The milky liquid found in all of these plants' organs is hazardous due to the presence of cardiac glycosides, such as neriifolin and peruvoside, which have been examined (Usman *et al.*, 2009).

The fruit has a milky juice that contains the heart stimulant thevetin. When left unchanged, all plant parts especially seeds are very dangerous. The green, lance-shaped leaves are arranged in a spiral pattern. They are linear and range in length from 13 to 15 cm. To stop water loss, the leaf surface is coated with wax. A deep red or black fruit with a yellow, trumpet-like flower encases a massive seed that looks like a Chinese "lucky nut." A deep red or black fruit with a yellow, trumpet-like flower encases a massive seed that looks like a Chinese "lucky nut." The stem changes from green to silver or grey as it ages. Root paste is used as a plaster to treat tumors. The plant thrives in shaded spots and a variety of well-irrigated soil types. Andre Thevet (1502–1590), a French missionary who collected species in South America, gave the plant the scientific name *Thevetia*. The seeds and sap of the yellow oleander are toxic. According to Usman *et al.* (2009), the seeds include a heart stimulant. It is beneficial for rehabilitating degraded land since it can flourish in harsh weather and supplies oil to Chinese industry. *Thevetia* cake has a 48% protein content after fat extraction, but Atteh *et al.* (1995)

state that it is hazardous for broilers and requires additional processing before it can be effectively used as a component of animal feed.

Two plant species that thrive in temperate regions around the world include *Thevetia peruviana*, also known as Nerium oleander, and yellow oleander, commonly known as common oleander. "Cardiac glycosides," which have effects similar to those of the heart drug digoxin, are found in both species. Both species can be dangerous when ingested orally and may possibly cause human death, according to Natural Standard Research Collaboration (2007). When grown as hedges, they can yield 400–800 fruits annually, depending on rainfall and plant age. The spherical, often green fruits gradually turn black as they mature. When a fruit is divided horizontally and diagonally, its shell may contain one to four seeds.



Figure 2:1: Thevetia peruviana plant

The green fruits have one to four seeds in their kernel and turn black as they develop. A longitudinally and transversely split nut is present in every fruit. Mature fruits contain about 22- 23% kernel and the seeds have 60-65% oil content (Deka and Basumatary, 2011). This plant reproduces by seed. Locally, it is also propagated by cuttings, especially for use as a live hedge.

All parts of the plant are toxic, due to the presence of glycosides. The toxicity of the glycoside is reflected in the accidental poisonings that occur among children that feed on the seed of the plants (Brewster 1986; Shaw and Pearn 1979). Some adults have reportedly died after consuming oleander leaves in herbal teas (Haynes *et al.*, 1985). According to Saravanapavanatha (1985), the kernel of about ten fruits may be fatal to an adult while kernel of one fruit may be fatal to children. Small children and cattle are typically more susceptible to *Thevetia peruviana* poisoning. Several workers have reported cases of livestock toxicity following the consumption of *Thevetia*. For example, Singh and Singh (2002) found that the plant's leaf, stem, and bark extracts killed fish. These extracts, when combined with seed kernel extract, also killed albino rats and caused poisoning symptoms (Oji and Okafor, 2000). After ten days of feeding, rats fed 20 and 30% of *Thevetia* seed kernels died at rates of 80 and 90%, according to Pahwa and Chartterjee (1990).

2.2.2 Taxonomical classification

1. Kingdom: Plantae
2. Subkingdom: Tracheobionta
3. Super division: Spermatophyta
4. Division: Magnoliophyta
5. Class: Magnoliopsida
6. Subclass: Asteridae
7. Order: Gentianales

8. Family: Apocynaceae
9. Genus: *Thevetia*
10. Species: *peruviana*

2.2.3 Chemical Composition of *Thevetia peruviana* Seed Oil

According to Usman et al. (2009), the cake of *Thevetia peruviana* has 30–37% protein, whereas the seed has a high oil content (60–65%). The physicochemical characteristics of *Thevetia peruviana* seed oil from three different places were examined by Ibiyemi et al. Free fatty acid, oil yield, saponification value, unsaponifiable matter, iodine value, peroxide value, refractive index, specific gravity, and viscosity (centipose) are among the parameters examined. Similarly, the amino acid composition of *Thevetia peruviana* seed cake was determined by Oluwaniyi et al. in 2007. In addition to saturated acids like palmitic (15%–24%) and stearic (10%–15%), it is mostly made up of unsaturated fatty acids, especially oleic acid (43%–64%) and linoleic acid (6%–20%).

2.2.4 Key Chemical Elements and Profile of Fatty Acids

With a favorable fatty acid composition of roughly 43.72–64% oleic acid (C18:1), 15.56–23.89% palmitic acid (C16:0), 6.3–19.79% linoleic acid (C18:2), and 10.71–15.98% stearic acid (C18:0), along with trace amounts of linolenic, arachidic, margaric, and behenic acids, *Thevetia peruviana* seed oil is a promising non-edible feedstock for industrial uses, especially the production of biodiesel.

The physicochemical properties of *Thevetia peruviana* seeds ranges from 56% to 64.7%, the saponification value is 182–191 mg KOH/g, the iodine value is 38.5–82 g I₂/100 g, the density is 0.89–0.91 g/cm³, and the free fatty acid content is 0.45% to 0.85%.

2.2.5 Toxicity and Uses

Despite its toxicity, *Thevetia peruviana* is used medicinally across the tropics. A decoction made from bark or leaves is used as an emetic, to relieve constipation, and to treat sporadic fevers. In Senegal, amenorrhea is treated using water that has been macerated with leaves and bark. Latex is used in Mali to soften calluses and corn. Leaf sap is used as eye and nose drops to treat severe headaches in Côte d'Ivoire and Benin. It is also applied to the nostrils to treat colds and revive persons who have passed out. The Luo people of Kenya treat colds with water that has been infused with crushed leaves.

A possible application for the seeds is as a purgative. In India, skin diseases are treated by using the seed oil externally (Pandya *et al.*, 2013; Usman *et al.*, 2009). Since toxic doses are just slightly greater than therapeutic ones, caution should be exercised in all medical applications, especially those used internally (Bandara *et al.*, 2010; Langford & Boor, 1996). A root infusion is used to treat snakebite injuries (Usman *et al.*, 2009) in Benin and Uganda. In Ghana, a decoction of the leaves is used to cure fever, jaundice, and intestinal worms. In addition to being used illegally, the bark and seeds are used to poison rats. The seeds are employed as an arrow or ordeal poison in Cameroon and southern Africa. Seeds have been used to kill or commit suicide in Sri Lanka and India. According to other sources, the seeds are used as an abortifacient. The seeds are employed as an insecticide when crushed with a soap solution, but they also function as a contact toxin.

The wood is used to produce building poles and tool handles in Ghana and Uganda. It serves as fuel as well. Sometimes they eat the fruit pulp. Stock do not graze the foliage, and the plants serve as a practical living fence. *Thevetia peruviana* is frequently planted as a hedge and as a decorative in gardens. It can be cultivated outdoors in the summer and in tubs in a glasshouse in colder climes. Additionally, it is grown to preserve soil or provide shade. The initially toxic seed oil can be consumed once it has been purified.

2.3 METHODS OF BIODIESEL PRODUCTION

Transesterification is the most used method of producing biodiesel. Figure 2.3 illustrates how triglycerides, the primary constituents of vegetable oils, react with an alcohol to form glycerol and fatty acid mono-alkyl esters (Ezeanyananso *et al.*, 2010). Because it is less expensive than other alcohols, methanol is the most widely used alcohol. The reaction in this instance is called methanolysis. Three mol of methanol and one mol of triglyceride are needed in the stoichiometry of the methanolysis reaction to produce three mol of fatty acid methyl ester and one mol of glycerol. Diglycerides and monoglycerides are formed in the middle of three successive reversible processes that make up this reaction.

Following the completion of the reaction, the glycerol is separated by centrifugation or settling, and the resulting layer is purified for use in either its more recent uses (animal feed, carbon feedstock in fermentations, polymers, surfactants, intermediates, and lubricants) or its more established uses (pharmaceutical, cosmetics, and food industries) (Claude, 1999). Prior to being utilized as diesel fuel, the biodiesel phase is additionally refined. Usually, a basic or acid catalyst is used to catalyze this reaction. However, because the process is quicker and the reaction conditions are controlled, base catalysts (KOH or NaOH) are the most widely utilized in the manufacture of biodiesel (Freedman *et al.*, 1984).

However, by neutralizing the free fatty acid in the oil and triglyceride saponification, their use in vegetable oil transesterification results in soaps. Because they partially burn the catalyst, reduce the yield of biodiesel, and complicate the separation and purification processes, soap forms are unwanted side reactions (Vicente *et al.*, 2004).

2.4 TRANSESTERIFICATION PROCESS

Triglyceride and methanol combine to form diglyceride in the first stage of the transesterification reaction (Freedman *et al.*, 1986; Bateni *et al.*, 2017; Orege *et al.*, 2022).

Additionally, diglyceride and methanol combine to form monoglyceride. Lastly, methanol and monoglyceride combine to form glycerol and methyl ester. Diglyceride and monoglyceride intermediate reactions could be disregarded if the process is assumed to be a one-step transesterification. Accordingly, three mol of methanol and one mol of triglycerides were needed for the transesterification reaction stoichiometrically. Conventional transesterification techniques, such as homogeneous, heterogeneous, enzyme, and advanced technology procedures (microwave, ultrasonic, or plasma assisted process), can be used to make biodiesel (Buchori *et al.*, 2016; Wang *et al.*, 2023). When compared to traditional techniques, plasma-assisted technology offers a number of benefits, including a faster reaction time, no soap or glycerol product, and a greater production of biodiesel (Purwanto *et al.*, 2020).

Transesterification, which uses lipid and low carbon alcohol as raw materials in the presence of a catalyst, is the most popular process for producing biodiesel. In the process of transesterification, the performance of the catalyst is the key factor of the biodiesel yield. The transesterification reaction involves the replacement of the alkyl group of an ester by another through interaction of the ester and alcohol (Cancela *et al.*, 2012; Marchetti & Errazu, 2008). Generally, this reaction is catalyzed by a base or an acid. The basic catalysts are the most common since the process is faster and the reaction conditions are moderated. The first step is a catalytic reaction with alcohol, producing an alkoxide. The alkyl ester is created when the alkoxide nucleophilically attacks the triglyceride's carbonyl group, forming a tetrahedral intermediate molecule. After the catalyst is deionized, the active chemical regenerates and can react with a fresh alcohol molecule to start a new catalytic cycle.

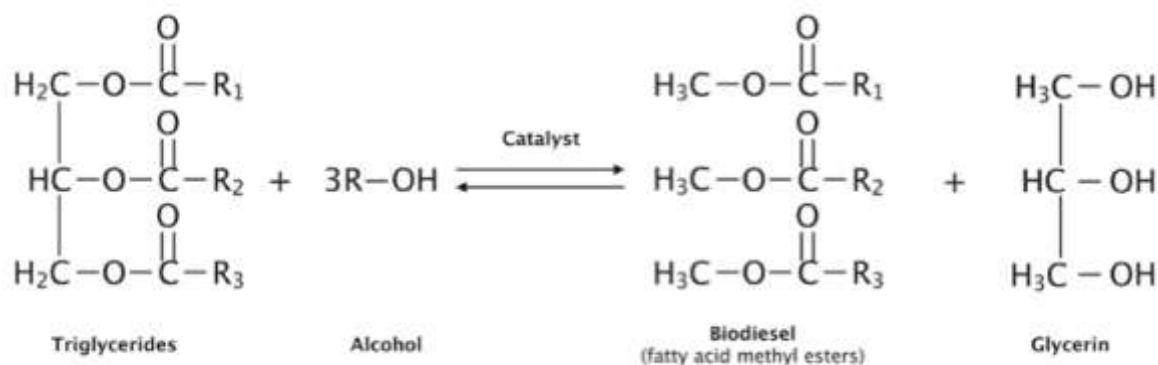


Figure 2:2: Biodiesel Production by Transesterification

2.5 CATALYSTS USED IN BIODIESEL PRODUCTION

Alkali catalysts, such as sodium hydroxide, sodium methoxide, potassium hydroxide, and potassium methoxide, are more effective than acidic or heterogeneous catalysts when it comes to transesterification of triglycerides (Ma and Hanna, 1999). The acids could be organic sulfonic acid, hydrochloric acid, phosphoric acid, or Sulphuric acid. Acid catalyzed transesterification is appropriate if the oil has more water and a high free fatty acid concentration. Using leftover vegetable oil, acid-catalyzed transesterification was investigated. In the presence of 100% excess alcohol, the reaction was carried out at four distinct catalyst concentrations: 0.5, 1.0, 1.5, and 2.25 M HCl. The outcome was compared with 2.25M H₂SO₄, and a decrease in viscosity was noted. In the concentration range of 1.5–2.25 M, H₂SO₄ exhibits superior catalytic activity (Mohamad and Ali, 2002).

Sodium hydroxide or potassium hydroxide, both in concentrations ranging from 0.4 to 2% w/w of oil, are typically utilized in the alkaline methanolysis process. Successful conversion was achieved using 1% sodium hydroxide or potassium hydroxide catalyst on refined and crude oils. The greatest yields and viscosities of the esters were obtained by methanolysis of soybean oil using 1% potassium hydroxide as a catalyst (Tomasevic and Marinkovic, 2003). Beef tallow

methanolysis was investigated using the catalysts NaOH and NaOMe. When comparing the two catalysts, NaOH outperformed NaOMe by a large margin (Ma et al., 1998). At 0.3 and 0.5% w/w of the beef tallow, respectively, the catalysts NaOH and NaOMe reached their maximum activity. Many byproducts, mostly sodium salts, are produced by sodium methoxide and should be disposed of as waste.

There have been attempts to produce fatty acid methyl esters by transesterifying rapeseed oil using basic alkaline-earth metal compounds. If there are methoxide ions in the reaction media, the reaction continues. Because the reaction mixture is a three-phase system oil-methanol-catalyst, which limits the reaction due to diffusion, the alkaline-earth metal hydroxides, alkoxides, and oxides catalyzed reaction occurs slowly (Gryglewicz, 1999). During the transesterification of rapeseed oil, the catalytic activity of magnesium oxide, calcium hydroxide, calcium oxide, calcium methoxide, barium hydroxide, and, for comparison, sodium hydroxide was examined. In this reaction, sodium hydroxide showed the greatest catalytic activity. After 30 minutes and 1.5 hours, respectively, the substrates' degree of reaction reached 85% and 95%, indicating a near equilibrium. After 30 minutes, the conversion of barium hydroxide was 75%, which was a little less active. The action of calcium methoxide was moderate. The substrates had reacted to a degree of 55% after 30 minutes, 80% after 1 hour, and 93% after 2.5 hours. When CaO was used as a catalyst, the rate of reaction was the slowest. In the methanolysis of rapeseed oil, neither calcium hydroxide nor magnesium oxide demonstrated any catalytic activity.

2.6 PROPERTIES OF BIODIESEL

There are two categories of characteristics that determine the quality of biodiesel. The other group specifically defines the chemical makeup and purity of fatty acid alkyl esters, whereas

the first group includes general parameters that are also utilized for fuel based on mineral oil (Mittelbach et al., 1996).

2.6.1 Viscosity

According to the Institute of Petroleum UK (IP, 1983), viscosity is the degree to which a fluid flow easily. It is a fluid's shear stress to shear rate ratio. When the fuel has the right viscosity, diesel engine fuel injection pumps and injectors work best. The actual spray won't pass through the combustion chamber and adequately mix with the air if the fuel isn't viscous enough. However, if the gasoline is too viscous, the fuel droplets will be too big to break up and mix with the air. Additionally, the relatively chilly cylinder walls may be struck by the heavy, viscous droplets and fail to ignite. As a result, viscosity has a significant impact on the fuel's atomization and penetration during injection. The degree of penetration required decreases with engine size. As a result, larger engines can handle more viscous gasoline than smaller engines. However, if viscosity is too low, more frequent maintenance of the protection system may be necessary, and if it is too high, the injection system may experience excessively high pressures (Clark, 1988).

Say Bold Universal (ASTM D88) and Kinematic (ASTM D445) are two widely used techniques for determining the viscosity of diesel fuel. Viscosity is typically 51 for all methods at 40 degrees Celsius. According to the ASTM D975 standard categorization of diesel fuel oil, the viscosities of the majority of grade 2 diesel fuels fall between 2.0 and 4.3 centistokes (32.6 and 40.1 say bolt seconds) at 40°C. At 40 degrees Celsius, the viscosity of grade 1 diesel fuels ranges from 1.4 to 1.9 centistokes.

2.6.2 Specific gravity

An indicator of a substance's density or weight per unit volume is called specific gravity. A liquid fuel's specific gravity (SG) is calculated by dividing its volume at a particular

temperature by the weight of an equivalent volume of fresh water at the same temperature. For both liquids, it is customary to specify specific gravities at either 15°C or 60°F, i.e., 15°C/15°C or 60°F/60°F. Density, which is measured in grammes per cubic centimetre (g/cm³) for solids, grammes per millilitre (g/ml) for liquids, and grammes per litre (g/l) for gases, is the quantity of a substance per unit volume. However, density and specific gravity actually have very little numerical difference, thus they can be considered interchangeable for all intents and purposes (Clark 1988).

2.6.3 Sulphur

Sulphur content varies depending on the source of crude oil, refining procedures, and fuel-grade diesel fuels. Higher boiling reactions typically contain more Sulphur. Sulphur in the fuel can have a negative impact on engine wear and deposit development, while the effects vary greatly and are mostly dependent on operating conditions. When operating diesel engines intermittently and at low temperatures, high Sulphur levels might cause issues. More moisture condensation may occur in these circumstances, and the fuel's Sulphur may react with the water to produce Sulphuric acid, which would exacerbate internal engine part wear and corrosion. The bomb technique (ASTM D129) is typically the basis for Sulphur standards. While heavier distillate and distillate residual fuel blends used in low and medium speed engines under sustained load, constant speed operation may have up to 2.0% Sulphur, diesel fuels for automotive service (trucks, buses, tractors) typically have Sulphur concentrations around 0.5% (weight). Sulphur concentration is trending downward due to air pollution concerns (Clark, 1988).

2.6.4 Cetane number

Diesel fuel's ignition quality is rated using the cetane number. Nhexadecane, or cetane, has a very good ignition quality (short ignition delay). On the cetane number scale, it stands for 100,

whereas the low-grade reference fuel, 2,2,4,4,6,8,8-heptamethylnonane (HMN), has a cetane number of 15. The phrase % cetane yields the cetane number. The fuel's hydrocarbon content and, to a lesser degree, its volatility characteristic are the main factors influencing the ignition delay duration, even if engine operating circumstances also have an impact. A fuel is considered to have poor ignition quality if it has a considerable delay period. The cetane number is used to express this fuel feature; the greater the value, the better the ignition quality.

There is a lag in the diesel combustion cycle between the beginning of fuel injection and the point at which burning releases enough energy to raise the cylinder's pressure over what can be achieved through air compression alone. An essential component of diesel engine combustion is this ignition delay time. When the accumulated fuel charge eventually begins to burn, pressure may build too quickly if the delay interval is too lengthy, resulting in knock or harsh engine operation (Clark, 1988).

2.6.5 Pour and cloud points

According to the Institute of Petroleum, an oil's cloud point is the temperature at which, when cooled under specific conditions, a cloud or haze initially forms (Clark, 1988). 3 degrees Celsius above the temperature at which oil simply stops flowing when chilled under specific circumstances is known as the pour point of oil. The cloud point is found using IP method 219, while the pour point is found using IP method 15. Physically speaking, all diesel fuels, particularly residual ones, include a small amount of paraffin wax in solution at room temperature. The cloud point is the temperature at which long, needle-like crystals of wax emerge from solution as the temperature drops. The interlacing crystals stop the remaining liquid fuel from flowing as the temperature drops even further. The pour point is here. When it comes to handling and storage in outdoor tanks and ship bunkers, the cloud and pour points are very crucial. In cars with diesel engines, where the engine is supplied by tiny pipes beneath the

chassis, cloud and pour points are equally crucial. When operating in cold weather, these exposed pipelines need to be effectively insulated to avoid fuel hunger. Although it is uncommon in warm engine rooms, wax production with paraffinic distillate fuels can clog small-bore engine fuel lines and filters in cold temperatures. However, it has proven problematic in some frigid climates outside of 52 compartments that have diesel-powered bow and stern thrusters and emergency diesel generators that are only occasionally used (Clark, 1988).

2.7 FACTORS AFFECTING BIODIESEL PRODUCTION

2.7.1 Temperature

The production of biodiesel will be significantly impacted by temperature. For instance, when oils become less viscous at higher temperatures, the rate of reaction increases and the reaction time decreases. However, because higher reaction temperatures speed up the saponification of triglycerides, Leung and Guo (2006) and Eevera et al. (2009) discovered that raising reaction temperatures over the ideal range results in a drop in biodiesel output. To stop alcohol from evaporating, the transesterification reaction temperature should typically be lower than the alcohol's boiling point. Depending on the oils or fats employed, the ideal reaction temperature might range from 50°C to 60°C (Leung and Guo, 2006; Ma and Hanna, 1999; Freedman et al., 1984).

2.7.2 Reaction time

The conversion of fatty acid esters increases as reaction time increases, according to Freedman et al. (1986). Because the alcohol and oil are mixed and dispersed, the reaction is first delayed. The reaction then continues very quickly. However, it took less than 90 minutes to reach the maximal ester conversion. The yield product, biodiesel/mono alkyl ester, does not increase

with additional reaction time (Leung and Guo, 2006; Alamu et al., 2007). Additionally, a longer reaction time reduces the final product (biodiesel) since transesterification is a reversible reaction that results in the loss of esters and the creation of soap (Eevera et al., 2009; Ma et al., 1998).

2.7.3 Catalyst concentration

Catalyst concentration has an impact on the production of biodiesel as well. The most popular catalysts for the generation of biodiesel are potassium hydroxide (KOH) or sodium hydroxide (NaOH). However, Freedman et al. (1984) discovered that sodium methoxide would be more successful because mixing sodium hydroxide with methanol produces very little water, which prevents the hydrolysis reaction from forming the final product (biodiesel) (Guo, 2005). This is one of the reasons catalyst is mixed with methanol before being applied to oil or fats. Additionally, the conversion of triglycerides into biodiesel increases as catalyst concentration increases with oil samples. Conversely, insufficient catalyst causes triglycerides to be partially converted into fatty acid esters (Leung and Guo, 2006; Guo, 2005). However, when the concentration of NaOH reached 1.5 weight percent, the best product yield (biodiesel) was obtained; however, additional increases in catalyst concentration were found to have a detrimental effect on the final product yield. Triglycerides react with extra alkali catalyst to produce additional soap (Leung and Guo, 2006; Gabelman and Hwang, 1999).

2.7.4 Alcohol-to-Oil Ratio

One of the key factors influencing the ester production is the molar ratio of alcohol to triglyceride. To produce three moles of fatty acid alkyl esters and one mole of glycerol, the stoichiometric ratio for transesterification calls for three moles of 45 alcohol and one mole of triglyceride. Transesterification, on the other hand, is an equilibrium reaction that requires a significant excess of alcohol to move the reaction to the right. A molar ratio of 6:1 is ideal for maximal conversion to the ester. The acid, peroxide, saponification, and iodine value of methyl

esters are unaffected by the molar ratio (Tomasevic and Marinkovic, 2003). However, due to an increase in solubility, the high molar ratio of alcohol to vegetable oil prevents glycerin from being separated. Glycerin lowers the yield of esters by driving the equilibrium back to the left when it is still in solution. Ethanol transesterification of cynara oil was investigated at molar ratios ranging from 3:1 to 15:1. Up to a value of 12, the ester yield rose as the molar ratio did. Molar ratios between 9:1 and 12:1 produced the best results. The reaction was not complete at molar ratios below 6:1. Because some of the glycerol is still in the biodiesel phase at a molar ratio of 15:1, it is challenging to separate the glycerin and the apparent yield of esters is reduced (Tomasevic and Marinkovic, 2003; Ezeanyananso et al., 2010).

2.8 ADVANTAGES AND DISADVANTAGES OF BIODIESEL

Emissions of carbon (IV) oxide, Sulphur (IV) oxide, carbon monoxide, and hydrocarbons are significantly decreased when using biodiesel. The percentage decrease in emissions with various blends of biodiesel is displayed in Figure 3. Any diesel engine may run on biodiesel, a renewable energy source, with little to no engine changes (Bilgin et al., 2015; Leidel, 2014; National Biodiesel Board, 2018). Biodiesel is simple to make and renewable. The raw materials are readily available, and the technique used in its production is not complicated. Biodiesel can be made from recovered cooking oil waste. Compared to fossil diesel, biodiesel has superior fuel qualities and is more biodegradable.

The manufacturing of biodiesel will draw investors to the area and assist create jobs for the unemployed. It may also be a source of foreign exchange revenues. *Jatropha* can be grown on low-grade, idle land to produce biodiesel. The production of biodiesel will boost rural economies and increase the amount of energy available to Africa's growing population. Competition with food could result from the manufacturing of biodiesel. More biodiesel is

produced from palm and soybean oils. This can be avoided by using other non-edible oils, like jatropha oil. The total cost of making biodiesel might be high; smart raw material selection can reduce this expense. Glycerine can be sold to businesses that require it, and the alcohol used in the process can be recovered and used again. There could be a shortage of land for food production. Oil crops that grow well on low-grade ground should be grown there. Pure biodiesel (B100) can have an impact on pump systems and cause maintenance issues. We are working to make sure that this is fixed. Diesel engines have successfully utilized B80 (National Biodiesel Board, 2018). Because of its high freezing point, biodiesel frequently causes issues when used in cold climates; blends as low as B5 should address this issue.

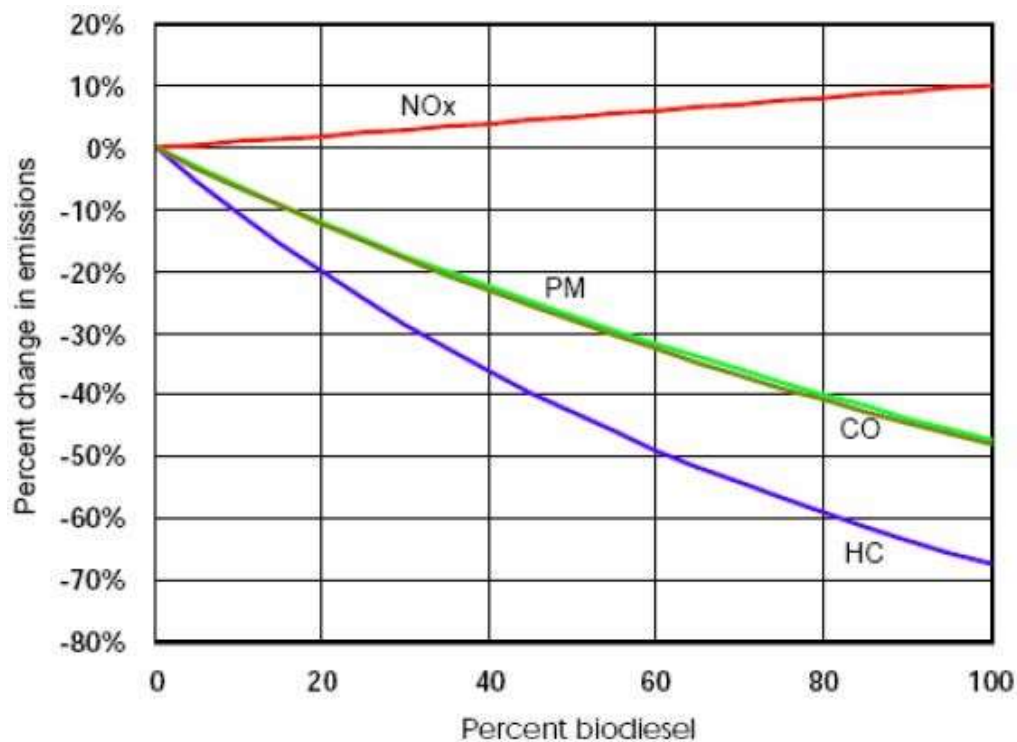


Figure 2:3: Biodiesel Blends and Their Emissions.

CHAPTER THREE

MATERIALS AND METHODS

3.1. MATERIALS

3.1.1 Raw Materials

The raw material used for this study is *Thevetia peruviana* oil.

3.1.2 Chemicals and Reagents

Methanol (CH_3OH) as the alcohol reactant, sodium hydroxide (NaOH) as the alkaline catalyst, KOH, acetic acid (CH_3COOH) for neutralization (vinegar), sodium sulfate, and distilled water for purification were among the chemical reagents utilized.

3.1.3 Apparatus Used

The apparatus used are vacuum pressure, separatory funnel, spatula, glass stoppers, three-neck round bottom flask, conical flask, beaker, thermometer, magnetic stirrer, silicone grease, bath wash, measuring scale, Liebig condenser, retort stand, pipette, beaker, measuring scale, thermometer, filter paper.

3.2. METHODS

A base-catalyzed transesterification technique was used to produce biodiesel.

3.2.1. Preparation of Biolubricant

Fill a three-neck round-bottom flask with 100g of FAME. Next, submerge the flask in silicon oil and set it on the heater. The flask is held in place by a retort stand, and it is securely held in place by an adaptor that is put into one of the necks using silicon grease. The flask and adaptor are linked to a Liebig condenser.

After heating the FAME to 70°C, insert a thermometer inside the flask to measure the temperature and install a stopper on the second neck. Pour 10g of TMP into the FAME solution after weighing it. Methanol is the byproduct of the reaction between TMP and FAME.

Trimethylolpropane (TMP) + 3 Fatty Acid Esters (FAME) = TMP Triester (Biolubricant) + 3 Methanol.



Figure 3:1: Preparation of Biolubricant

3.2.2. Preparation of Methoxide Solution

Weighing 5g of methanol into 0.3g of KOH in a flask resulted in a methoxide solution. Gently stir the mixture until all of the KOH has dissolved completely. This reaction forms potassium methoxide and water. Potassium methoxide is commonly used as the catalyst for transesterification of oil and fats with methanol.

Transfer the KOH and methanol solution into a dropping funnel, then cover the mouth with silicon grease to secure it to the final round-bottom flask with three necks.

Once dropping funnel is inserted, turn on the tap and let it gradually descend before raising the heat and temperature for 4 hours.

After heating, let it cool before adding a small amount of ethyl acetate and filtering to obtain the TMP.

3.2.3. Preparation of Biodiesel via Transesterification

A round-bottom flask was filled with 100g of *Thevetia* oil. The flask was submerged in a 60°C water bath. A magnetic stirrer is used to maintain a steady temperature in the water bath.

A beaker was filled with the 15.82g of methanol. After weighing 1g of NaOH, it was added to the measured methanol. Continue stirring the NaOH until it dissolves in the methanol. Methanol and NaOH react in an exothermic manner.

Pour into the oil after dissolving the NaOH in the methanol, keep the temperature between 55°C and 60°C for 2 hours, and then place the condenser above the conical flask. The colour changes to Dark Brown.

3.2.4. Separation of Biodiesel

After heating the mixture for 2 hours, take out the condenser, add 2 drops of acetic acid (vinegar), and turn off the stirrer. To separate, pour the mixture into a separating funnel. You have two layers after adding a separating layer, which are

- The FAME (Biodiesel) is the top layer.
- Glycerol is the bottom layer.



Figure 3:2: Separation of Biodiesel

3.2.5. Purification of Biodiesel

After draining the glycerol from the funnel's bottom, it was weighed. The separatory funnel was filled with distilled water heated to 50°C in order to cleanse the leftover biodiesel. After giving the mixture a good shake, it was left to settle. The cloudy, contaminated water was drained from the bottom. This washing process was carried out multiple times until the discarded water was crystal clear, signifying the elimination of unreacted methanol and leftover catalyst.

A flask was filled with 20g of anhydrous sodium sulphate (Na_2SO_4) in order to eliminate any last traces of moisture from the biodiesel. After that, the cleaned biodiesel (FAME) was poured

into the flask with the drying agent and constantly shaken. This made sure that all of the remaining water molecules were absorbed. A clear, pure biodiesel product was obtained by filtering the mixture using a funnel and filter paper to get rid of the hydrated sodium sulphate and any other suspended particles.

3.2.6. Determination of the Yield

The weight of the recovered glycerol was subtracted from the starting quantity of the oil used to calculate the biodiesel yield.

To Calculate the mass of biodiesel;

By using the data given,

Mass of Oil = 100g

Weight of Glycerol = 17.91g

$$\text{Mass of Biodiesel} = \text{Mass of Oil (g)} - \text{Weight of the Glycerol (g)}$$

$$\text{Mass of Biodiesel} = 100\text{g} - 17.91\text{g}$$

$$\text{Mass of Biodiesel} = 82.09\text{g}$$

The percentage of yield is calculated as;

$$\text{Yield \%} = \frac{\text{Actual Mass of Biodiesel}}{\text{Initial Mass of Oil}} \times 100$$

$$\text{Yield \%} = \frac{82.09}{100} \times 100$$

$$\text{Yield} = 82.09\%$$

3.2.7. Fourier Transform Infrared Spectroscopy (FTIR) of Biodiesel

FTIR analysis is used to examine the structural changes and thermal stability of a polymeric, organic, and inorganic material. A chemical analysis of biodiesel made from *Thevetia*

peruviana oil seed was finished using FTIR. We added a tiny bit of biodiesel to the machine and watched for the result to appear in order to help analyze the test's outcome.

CHAPTER FOUR

RESULTS AND DISCUSSION

4.1. FOURIER TRANSFORM INFRARED SPECTROSCOPY (FTIR) ANALYSIS FOR BIODIESEL PRODUCED USING NaOH

Fourier Transform Infrared (FTIR) spectroscopy was used to determine the functional groups in the biodiesel made from *Thevetia peruviana* seed oil using NaOH catalyst. The related functional group assignments are displayed in Table 4.1, and the resulting spectrum is displayed in Figure 4.1.

At 3023.65, 2933.47, 2859.13, 1261.19, 953.43, 863.41, 652.36, and 443.37 cm^{-1} , the FTIR spectra showed distinctive absorption bands. The existence of unsaturated fatty acid chains is shown by the absorption band at 3023.65 cm^{-1} , which is associated with the stretching vibration of unsaturated ($=\text{C}-\text{H}$) bonds. The lengthy hydrocarbon chains typical of fatty acid methyl esters (FAME) are confirmed by strong absorption peaks at 2933.47 cm^{-1} and 2859.13 cm^{-1} , which are attributed to the asymmetric and symmetric stretching vibrations of aliphatic C-H bonds (CH_2 and CH_3).

One of the distinctive peaks of biodiesel made via transesterification is the absorption peak at 1261.19 cm^{-1} , which is ascribed to the C-O stretching vibration of ester functional groups. The effective conversion of triglycerides into fatty acid methyl esters is confirmed by the development of this ester peak.

Additionally, the peaks at 953.43 cm^{-1} and 863.41 cm^{-1} are linked to $=\text{C}-\text{H}$ bond out-of-plane bending vibrations, suggesting that the biodiesel contains unsaturated carbon-carbon double bonds. While the weak absorption at 443.37 cm^{-1} is linked to skeletal vibrations within the

molecular framework, the absorption band at 652.36 cm^{-1} is likewise located within the fingerprint region and is linked to alkene C–H bending vibrations.

Overall, the effective synthesis of biodiesel is confirmed by the presence of strong aliphatic C–H stretching bands and the ester C–O stretching vibration. These functional groups match the distinctive FTIR spectrum of fatty acid methyl esters found in earlier research, suggesting that *Thevetia peruviana* seed oil was successfully transformed into biodiesel by the transesterification process.

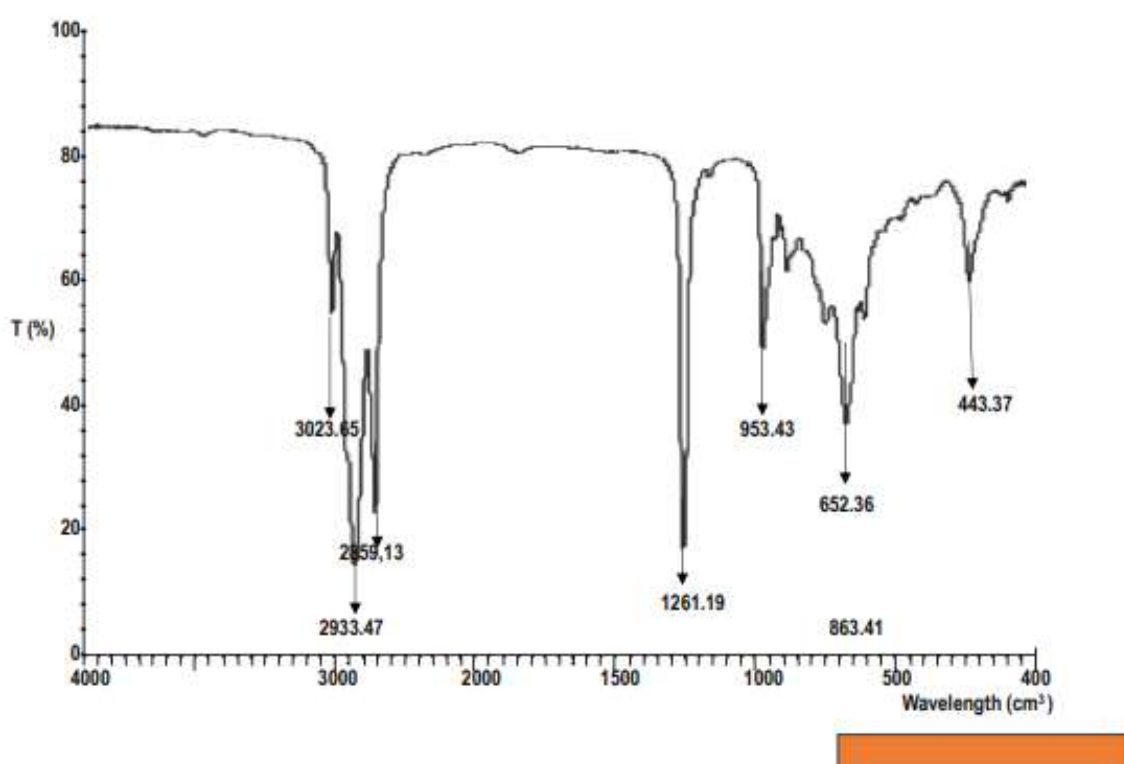


Figure 4:1:FTIR spectrum of *Thevetia peruviana* biodiesel using NaOH

Table 4.1: Functional groups in *Thevetia peruviana* biodiesel

Wavenumber (cm ⁻¹)	Functional group assignment
3023.65	=C–H stretching (alkene)
2933.47	C–H asymmetric stretching (alkane)
2859.13	C–H symmetric stretching (alkane)
1261.19	C–O stretching (ester)
953.43	=C–H bending (alkene)
863.41	C=C–H out-of-plane bending
652.36	C–H bending (alkene)
443.37	Skeletal vibration (fingerprint region)

4.2. FOURIER TRANSFORM INFRARED SPECTROSCOPY (FTIR) ANALYSIS OF BIODIESLE PRODUCED USING KOH

Figure 4.2 displays the FTIR spectrum of the biodiesel made from *Thevetia peruviana* seed oil using a potassium hydroxide (KOH) catalyst, and Table 4.2 displays the matching functional group assignments. The presence of long-chain hydrocarbon chains typical of fatty acid methyl esters (FAME) was indicated by the spectrum's distinctive absorption bands at 2922.2 cm⁻¹ and 2855.1 cm⁻¹, which are attributed to the asymmetric and symmetric stretching vibrations of aliphatic C–H bonds. The creation of methyl esters during the transesterification process is confirmed by a significant absorption band at 1744.4 cm⁻¹, which corresponds to the ester carbonyl (C=O) stretching vibration.

The bending vibration of terminal alkane C–H bonds is responsible for the absorption band at 1438.8 cm⁻¹, whereas C–O stretching linked to carboxylic functional groups is responsible for

the band at 1300.8 cm^{-1} . The C–O stretching vibration of ester groups is represented by the peak at 1170.4 cm^{-1} , while the C–O–C stretching vibrations typical of alkanoate esters are attributed to the absorption at 1013.8 cm^{-1} . The out-of-plane C–H bending vibrations of unsaturated hydrocarbons (alkenes) are responsible for the additional peaks seen at 849.8 cm^{-1} and 723.1 cm^{-1} . Strong ester carbonyl (C=O), ester C–O, and aliphatic C–H stretching vibrations demonstrate that *Thevetia peruviana* seed oil's triglycerides were effectively transformed into fatty acid methyl esters. When potassium hydroxide is used as the transesterification catalyst, these distinctive absorption bands show successful biodiesel generation.

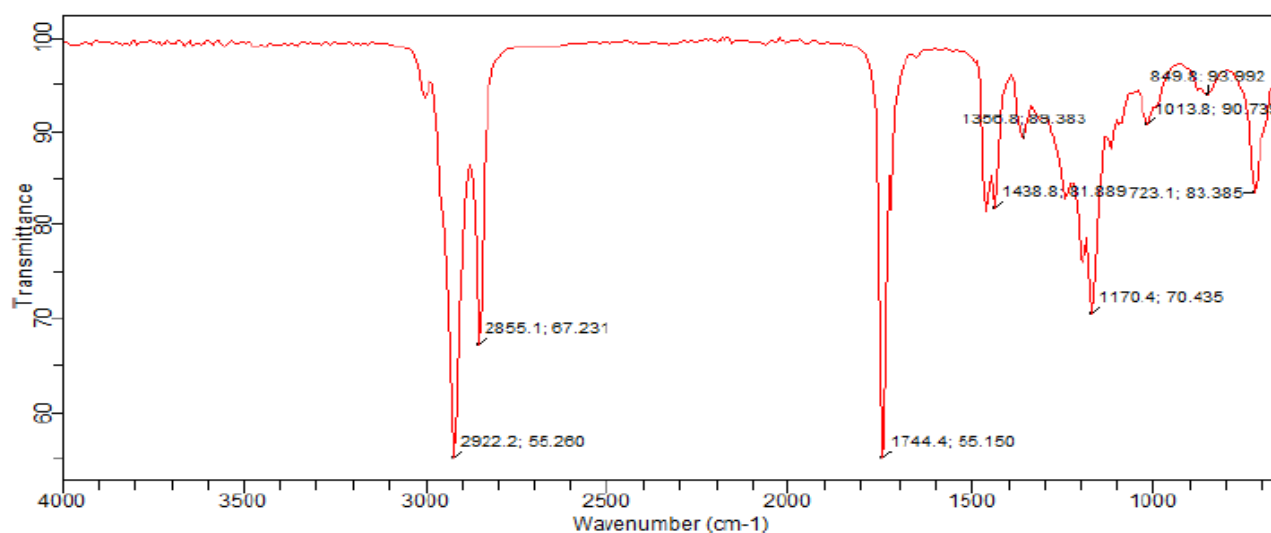


Figure 4:2: FTIR spectrum of *Thevetia peruviana* biodiesel using KOH

Table 4.2: Functional groups found in *Thevetia peruviana* biodiesel made with a KOH catalyst

Wavenumber (cm ⁻¹)	Functional Group Assignment
2922.2, 2855.1	C–H stretching (alkanes)
1744.4	C=O stretching (ester carbonyl)
1438.8	C–H bending (terminal alkane)
1300.8	C–O stretching (carboxylic group)
1170.4	C–O stretching (ester)
1013.8	C–O–C stretching (alkanoate ester)
849.8, 723.1	=C–H bending (alkenes)

4.3. FOURIER TRANSFORM INFRARED SPECTROSCOPY (FTIR) ANALYSIS *THEVETIA PERUVIANA* SEED OIL TVPO OIL

Figure 4.3 displays the crude *Thevetia peruviana* seed oil (TVPO) FTIR spectrum, and Table 4.3 displays the matching functional group assignments. The existence of long-chain hydrocarbon chains typical of triglycerides and fatty acids was indicated by the spectrum's distinctive absorption bands at 2922.2 cm⁻¹ and 2855.1 cm⁻¹, which are due to the asymmetric and symmetric stretching vibrations of aliphatic C–H bonds.

The ester carbonyl (C=O) stretching vibration was detected as a prominent absorption band at 1744.4 cm⁻¹. This peak, which is typical of triglycerides, indicates that esterified fatty acids make up the majority of the oil. The C–H bending vibrations of aliphatic hydrocarbon chains are responsible for the absorption band at 1401.1 cm⁻¹. C–O stretching vibrations of ester functional groups are represented by the band at 1237.5 cm⁻¹, whereas C–O–C stretching vibrations linked to ester linkages in triglycerides are responsible for the peak at 1162.9 cm⁻¹. Other absorption bands were detected at 723.1 cm⁻¹ and 1071.7 cm⁻¹. The absorption at 723.1

cm^{-1} is associated with out-of-plane C–H bending vibrations of unsaturated hydrocarbons and long-chain methylene groups, whereas the peak at 1071.7 cm^{-1} is attributed to C–O stretching vibrations.

Strong ester carbonyl (C=O), ester C–O, and aliphatic C–H stretching vibrations verify that the extracted oil is mostly composed of long-chain fatty acid-rich triglycerides. These distinctive functional groups, which are common to vegetable oils, show that *Thevetia peruviana* seed oil is an appropriate feedstock for transesterification-based biodiesel generation.

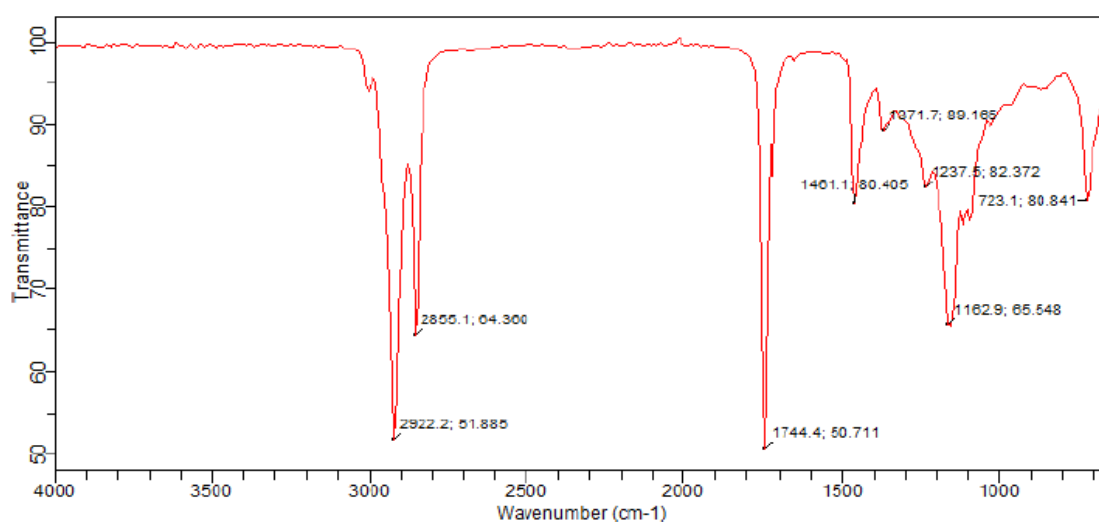


Figure 4:3: FTIR spectrum of *Thevetia peruviana* seed oil (TVPO)

Table 4.3: *Thevetia peruviana* seed oil's (TVPO) functional groupings

Wavenumber (cm ⁻¹)	Functional Group Assignment
2922.2, 2855.1	C–H stretching (alkanes)
1744.4	C=O stretching (ester carbonyl of triglycerides)
1401.1	C–H bending (alkanes)
1237.5	C–O stretching (ester)
1162.9	C–O–C stretching (ester linkage)
1071.7	C–O stretching (ester/alcohol)
723.1	=C–H bending / CH ₂ rocking (long-chain hydrocarbons)

4.4. FTIR SPECTRAL OF TVPO OIL, KOH BIODIESEL, AND NaOH BIODIESEL

To assess the efficiency of the transesterification process, the FTIR spectra of raw *Thevetia peruviana* oil (TVPO), biodiesel made with KOH catalyst, and biodiesel made with NaOH catalyst were compared. Table 4.4 summarizes the main absorption bands seen in the three samples.

Table 4.4: FTIR Absorption Band Comparison of Raw TVPO Oil, KOH-Produced Biodiesel, and NaOH-Produced Biodiesel

Functional Group	Raw TVPO Oil (cm ⁻¹)	TVPO KOH Biodiesel (cm ⁻¹)	TVPO NaOH Biodiesel (cm ⁻¹)	Interpretation
=C–H Stretching (Unsaturated alkene)	—	—	3023.65	Indicates unsaturation in fatty acid chains.
C–H Asymmetric Stretching (CH ₂)	2922.2	2922.2	2933.47	Present in all samples; confirms long hydrocarbon chains.
C–H Symmetric Stretching (CH ₃ /CH ₂)	2855.1	2855.1	2859.13	Characteristic of triglycerides and methyl esters.
Ester Carbonyl (C=O) Stretching	1744.4	1744.4	Not clearly observed	Strong indicator of ester compounds; clearly defined in TVPO and KOH biodiesel.
C–H Bending	1401.1	1438.8	—	Indicates aliphatic hydrocarbon chains.
C–O Stretching (Ester)	1237.5	1170.4	1261.19	Confirms ester functional groups produced during transesterification.
C–O–C Stretching	1162.9, 1071.7	1013.8	—	Characteristic of ester linkages.
=C–H Out-of-plane Bending	723.1	849.8, 723.1	953.43, 863.41, 652.36	Indicates the presence of unsaturated fatty acid chains.
Fingerprint Region	—	—	443.37	Skeletal molecular vibration.

The raw TVPO oil's FTIR spectra revealed the distinctive characteristics of triglycerides, such as a pronounced ester carbonyl ($\text{C}=\text{O}$) absorption at 1744.4 cm^{-1} and strong aliphatic $\text{C}-\text{H}$ stretching bands at 2922.2 and 2855.1 cm^{-1} . Additional $\text{C}-\text{O}$ and $\text{C}-\text{O}-\text{C}$ stretching bands verified that the vegetable oil was esterified.

The biodiesel spectra maintained the strong ester carbonyl peak at 1744.4 cm^{-1} along with prominent $\text{C}-\text{O}$ stretching bands at 1170.4 and 1013.8 cm^{-1} following transesterification using KOH catalyst. These peaks show efficient triglyceride conversion to biodiesel and are typical of fatty acid methyl esters (FAME). Additionally, the spectrum showed alkene bending bands at 849.8 and 723.1 cm^{-1} , indicating that the biodiesel still included some unsaturated fatty acid structures.

On the other hand, strong aliphatic $\text{C}-\text{H}$ stretching bands at 2933.47 and 2859.13 cm^{-1} and a clear ester $\text{C}-\text{O}$ stretching band at 1261.19 cm^{-1} were seen in the biodiesel made using the NaOH catalyst, confirming the synthesis of biodiesel. However, the NaOH spectra did not clearly show an ester carbonyl ($\text{C}=\text{O}$) absorption around $1740\text{--}1750\text{ cm}^{-1}$. While the NaOH biodiesel showed multiple alkene-related bending vibrations (953.43 , 863.41 , and 652.36 cm^{-1}), the evidence for methyl ester production was not as strong as it was in the KOH biodiesel spectrum due to the lack of a distinct carbonyl peak.

Because it included both the strong ester carbonyl absorption at 1744.4 cm^{-1} and the ester $\text{C}-\text{O}$ stretching bands that are commonly associated with fatty acid methyl esters, the KOH biodiesel displayed the most comprehensive collection of distinctive biodiesel peaks when the three spectra were compared. The ester $\text{C}-\text{O}$ and aliphatic $\text{C}-\text{H}$ vibrations in the NaOH biodiesel's spectra demonstrated successful transesterification as well, although there was no distinct carbonyl peak.

Hence, the raw TVPO oil retained the spectral characteristics typical of unconverted triglycerides, whereas the biodiesel produced using KOH catalyst showed better-defined biodiesel peaks and seemed to have undergone a more thorough conversion than the biodiesel produced using NaOH catalyst, according to FTIR evidence alone.

Table 4.5: Comparative Evaluation of FTIR Features.

FTIR Feature	Raw TVPO Oil	KOH Biodiesel	NaOH Biodiesel
Long-chain hydrocarbon (C–H stretching)	✓	✓	✓
Ester carbonyl (C=O) clearly observed	✓	✓	✗
Ester C–O stretching	✓	✓	✓
C–O–C ester linkage	✓	✓	Not prominent
Alkene (unsaturation) bands	✓	✓	✓
Overall evidence of biodiesel (FAME) formation	Feedstock (Triglycerides)	Strong	Moderate
Quality of FTIR spectrum	Feedstock spectrum	Excellent	Good

The existence of long-chain fatty acids was confirmed by the significant aliphatic C–H stretching vibrations seen in all three samples (Table 4.4). However, because it displayed both the distinctive ester carbonyl absorption at 1744.4 cm^{-1} and the ester C–O stretching bands linked to fatty acid methyl esters, the biodiesel made with KOH had the most comprehensive FTIR profile. A carbonyl peak was also seen in the raw TVPO oil at 1744.4 cm^{-1} , although this is thought to be caused by triglyceride ester links rather than methyl esters. The biodiesel made

with NaOH, on the other hand, showed noticeable aliphatic C–H stretching and ester C–O stretching vibrations but lacked a distinct ester carbonyl peak, which made the evidence of methyl ester generation relatively weaker. According to the FTIR spectra, the KOH catalyst produced biodiesel with more distinctive FAME peaks than the NaOH catalyst, suggesting a more thorough transesterification under the used circumstances.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1. CONCLUSION

This study used sodium hydroxide (NaOH) as a catalyst in the transesterification process to successfully produce biodiesel from *Thevetia peruviana* seed oil. Because of its high oil content and favorable fatty acid composition, the extracted oil's physicochemical properties validated its potential as a non-edible feedstock for biodiesel generation.

The source oil and the resulting biodiesel were characterized using Fourier Transform Infrared (FTIR) spectroscopy. The raw *Thevetia peruviana* oil's FTIR spectra showed distinctive absorption bands that corresponded to triglycerides, such as ester carbonyl (C=O) functional groups and aliphatic C–H stretching. The successful conversion of the oil into biodiesel was confirmed by the presence of distinctive absorption bands linked to fatty acid methyl esters (FAME) in the biodiesel made with NaOH catalyst after transesterification, especially the aliphatic C–H stretching and ester C–O stretching vibrations.

The FTIR spectra of the raw oil, NaOH-produced biodiesel, and KOH-produced biodiesel showed good agreement with the typical functional groups reported for biodiesel when compared to earlier research. The NaOH-catalyzed biodiesel likewise displayed the crucial functional groups needed to verify effective transesterification, even though the KOH-catalyzed biodiesel had a more noticeable ester carbonyl (C=O) absorption band. The catalyst type, reaction circumstances, purification process, and instrumentation variances could all be responsible for the minor variations in absorption bands.

The study's overall findings show that *Thevetia peruviana* seed oil is a suitable non-edible feedstock for the manufacturing of biodiesel. Its use lessens competition with food resources,

provides a sustainable substitute for edible oils, and advances the creation of ecologically friendly and renewable energy sources.

5.2. RECOMMENDATIONS

It is recommended that:

- To find the most effective and ecologically friendly catalyst for producing biodiesel from *Thevetia peruviana* seed oil, comparative research involving homogeneous and heterogeneous catalysts should be carried out.
- To evaluate *Thevetia peruviana* biodiesel's appropriateness for commercial diesel engines, tests on engine performance and exhaust emissions should be carried out using various blends of the fuel with regular diesel.
- *Thevetia peruviana* is an easily available, non-edible plant that should be cultivated on marginal or degraded soils in order to provide a sustainable source of feedstock for biodiesel without interfering with food production.
- More research should look into ways to apply appropriate antioxidants or blending techniques to enhance the oxidative stability and storage qualities of biodiesel made from *Thevetia peruviana* seed oil.
- The manufacture of biodiesel into a profitable business, more thorough research is needed to explore many useful by-products from glycerol and make appropriate use of oil cake.

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